

RUTUJA DESHMUKH

Aspiring AI Engineer | Building AI Applications for Real-World Problems

Nashik, Maharashtra, India | 9960310522 | rutujadeshmukh634@gmail.com

Portfolio: <https://rutuja-portfolio-two.vercel.app> | GitHub: <https://github.com/RutujaDeshmukh29> | LinkedIn:

<https://www.linkedin.com/in/rutuja-deshmukh29>

PROFESSIONAL SUMMARY

B.E. Artificial Intelligence & Data Science student (2nd year) with a 3-year Diploma in AI & ML. Builds and deploys practical AI applications with Python, LLMs, RAG and multi-agent workflows, using FastAPI and Next.js. Experience across personal projects, a Python/Django internship, a 24-hour hackathon, research presentations and a client-based engineering project. Seeking internships, entry-level AI/ML roles and freelance projects in Generative AI and AI application development.

TECHNICAL SKILLS

AI / Generative AI: Generative AI, RAG, LLM Integration, LangChain, LangGraph, Multi-Agent Systems, AI Agents, Prompt Engineering, Semantic Search, Vector Embeddings, Conversational AI

Programming and Backend: Python, SQL, JavaScript, FastAPI, Django, REST APIs

Databases: PostgreSQL, Supabase, pgvector, SQLite, ChromaDB

Frontend: Next.js, React.js, Streamlit, Tailwind CSS

Tools and Platforms: Git, GitHub, VS Code, Hugging Face, Groq API, Gemini API

EXPERIENCE

Python Application Development & Data Engineering Intern

June 2025 – August 2025

ProWorld Technology Pvt. Ltd.

- Built Django-based web applications, including UI development with HTML, CSS and Django templates.
- Worked on the AI-Based Energy Monitoring System and the EcoCart e-commerce website as part of a development team.
- Gained hands-on experience in data handling, testing, usability, debugging and team development workflows.

Client-Based Project: EngineerX (Ongoing)

AnaVruksha | Private repository

- Working on an AI-powered engineering workspace that applies agent workflows to real-world hardware engineering problems; V1 focuses on rover motor-control and power subsystems.
- Responsible for understanding client requirements, communicating technical ideas, coordinating tasks and managing assigned project responsibilities.
- Exploring AI agents, LLMs and engineering tools such as Fusion 360, KiCad, Gazebo and Isaac Sim.

PROJECTS

AI Multi-Agent Research & Engineering Workspace

Tech: Next.js, FastAPI, LangGraph, LangChain, Groq / LLMs, PostgreSQL, Supabase pgvector, Zustand, SSE streaming, Framer Motion

GitHub: <https://github.com/RutujaDeshmukh29/AI-Multi-Agent-Research-Engineering-Workspace>

- Built and deployed a multi-agent workspace with Research, Engineering, Planner and Critic agents and a QA Controller, orchestrated with LangGraph and LangChain.
- Implemented task/relevance routing and session memory through a FastAPI backend with PostgreSQL / Supabase pgvector.
- Developed a SaaS-style Next.js interface with real-time SSE streaming, Zustand state management and Framer Motion animations.

AdaptEd AI: Adaptive Learning Platform

Tech: Next.js, TypeScript, FastAPI, PostgreSQL, Supabase, ChromaDB, PyMuPDF, Sentence Transformers, Google Gemini API

GitHub: <https://github.com/RutujaDeshmukh29/Adaptive-AI>

- Team project (4 members) started at the Tech Fusion 24H Hackathon 2026 (Smart Education); my role was backend development and AI integration.
- Tracks topic mastery to build a learner model and gives explainable next-step recommendations, with adaptive quizzes and a prerequisite-based learning path.
- Includes a RAG tutor over students' uploaded PDFs with source citations (PyMuPDF, ChromaDB, Sentence Transformers).

AI Learning Companion

Tech: Python, Streamlit, ChromaDB, PyMuPDF, Sentence Transformers, Groq, Vector Embeddings

Live Demo: <https://ai-learning-companion-study-assistanti.streamlit.app/>

GitHub: <https://github.com/RutujaDeshmukh29/AI-Learning-Companion-Study-Assistant>

- Built and deployed a RAG-based study assistant: upload PDFs and chat with them using a hybrid RAG pipeline.
- Added summaries, quiz generation, coding and interview questions, roadmap tracking, weak-topic analytics, five learning modes, a voice assistant and Mermaid.js diagrams.

AgroBot: AI-Powered Smart Robotic Lawn Mower (Final-Year Diploma Project)

Tech: Python, Raspberry Pi, Node.js, IoT, Android integration

GitHub: <https://github.com/RutujaDeshmukh29/AI-Powered-Smart-Robotic-Lawn-Mower>

Android App: <https://github.com/RutujaDeshmukh29/AgroBot-Mower-Android-App>

- Contributed to software development and Raspberry Pi integration for an autonomous mower with GPS/maps, mission planning, remote control, obstacle alerts and mission history.
- Presented at technical competitions and exhibitions, including IET Expo 2026; the related research paper was published in IJARCCCE.

Other Projects

- **SignalIQ:** market and competitive-intelligence dashboard with signal scoring, competitor radar and a feedback-weighted recommendation engine (React, Vite, Recharts; seed data, live sources planned).
<https://github.com/RutujaDeshmukh29/SignalIQ>
- **Scriptora:** collaborative script-writing and project-management workspace; commissioned project in active development (Next.js, FastAPI, PostgreSQL). <https://github.com/RutujaDeshmukh29/Scriptora>
- **EcoCart:** Django e-commerce application with product catalogue, cart and order flow.
<https://github.com/RutujaDeshmukh29/EcoCart>

EDUCATION

B.E. Artificial Intelligence & Data Science

Expected 2029

Mumbai Educational Trust (MET), MET League of Colleges | Currently in 2nd year

Diploma in Artificial Intelligence & Machine Learning

Completed 2026

K. K. Wagh Polytechnic, Nashik | 3-year programme | 89.76%

Secondary School Certificate (SSC)

May 2023

Shree Ram Vidhyalaya, Nashik | 92.80%

RESEARCH, HACKATHONS AND ACHIEVEMENTS

- **Research Publication:** AgroBot research paper published in IJARCCCE (International Journal of Advanced Research in Computer and Communication Engineering). <https://ijarcce.com/papers/agrobot-ai-powered-smart-robotic-lawn-mower/>
- **IET Expo 2026:** presented AgroBot.
- **Avishkar Research Competition 2026–27** (Engineering & Technology): presented the research idea “AI-Based Early Prediction of Food Spoilage Risk Using Environmental and Storage Parameters Without Specialized Hardware.”
- **Tech Fusion 24H Hackathon 2026** (MET Bhujbal Knowledge City, 17–18 Sept 2026): participated in a 4-member team on Smart Education (Personalized AI Learning Assistant); backend development and AI integration.
- **First Prize** in the 2026 debate competition at MET Institute of Engineering.

CERTIFICATIONS

- Introduction to Generative AI – Google Cloud & Simplilearn
- Build AI Agents using LangGraph – Simplilearn
- AWS Academy Graduate, Cloud Foundations – Amazon Web Services
- Big Data Analytics Projects with Apache Spark – Infosys Springboard
- Database Management System – Infosys Springboard
- Python Training – Spoken Tutorial, IIT Bombay